

Convolution as a local operation

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Image tensors and channels

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and validation curves

Without opening the old lesson, explain one key idea from Training and validation curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Separate parameters and operations for Convolution as a local operation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Convolution as a local operation. Proposed explanation: “Each window has its own set of learned weights”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Filter a new image independently

Use image rows [2, 1, 0], [1, 3, 2] and kernel rows [0, 1], [-1, 2]. Use no padding, stride 1 and bias 0. Give the output shape, both four-product sums and the feature map. Then add a shared bias of -5 and apply ReLU.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace a local sum

Use the lesson image and kernel at window start [0, 1]. List the window rows, four products and output before activation.

My reasoning and evidence

6. W2 Separate parameters and operations

How many shared kernel weights does the lesson use? How many products are computed over all four valid windows? What changes if one shared bias is added?

My reasoning and evidence

7. W3 Check stride and padding

For the lesson image and kernel, give the output shape with no padding and stride 2. Then give the shape with one zero border on each side and stride 1, and calculate that padded map's top-left output.

My reasoning and evidence

8. W4 Filter a new image independently

Use image rows [2, 1, 0], [1, 3, 2] and kernel rows [0, 1], [-1, 2]. Use no padding, stride 1 and bias 0. Give the output shape, both four-product sums and the feature map. Then add a shared bias of -5 and apply ReLU.

My reasoning and evidence

9. W5 Test a convention error

A program reverses both axes of the lesson kernel to $[[1, 0], [-1, 1]]$. Compute the top-left output. Does it agree with our declared convention? Explain what the test checks.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled convolution as a local operation evidence sheet to Image-pattern Network Investigation. Show how you can compute a local weighted sum at each valid image position. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Convolution as a local operation

Draw the exact patch-by-patch arithmetic of a local image operation.

Image rows [1, 2, 3],[4, 5, 6]; kernel rows [1, 0],[0,-1]. Use unflipped kernel, no padding, stride1, bias0.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Image used by both local kernel windows

	col0	col1	col2
row0	1	2	3
row1	4	5	6

Read a cell using both its row and column labels.

Convolution as a local operation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First patch

$$1 \times 1 + 2 \times 0 + 4 \times 0 + 5 \times (-1) = -4.$$



2. Second patch

$$2 \times 1 + 3 \times 0 + 5 \times 0 + 6 \times (-1) = -4.$$



3. Feature map

Output shape 1 x 2 and map $\begin{bmatrix} -4 & -4 \end{bmatrix}$; the same kernel is reused at each position.

CHANGE ONE THING

Add shared bias2 after each patch sum.

Explain the changed result using the matching lesson.

Why is flipping the kernel an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Draw the exact patch-by-patch arithmetic of a local image operation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Output shape 1×2 and map $[-4, -4]$; the same kernel is reused at each position.

1. Feature map
2. First patch
3. Second patch

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Feature map Output shape 1×2 and map $[-4, -4]$; the same kernel is reused at each position. A local kernel combines the values in one small image window with a fixed set of weights. Slide that same kernel to the next valid window and repeat the products. Sharing weights lets the operation look for the same local pattern at different positions. This example declares an unflipped kernel; adding bias changes each window result without changing which pixels were used.

Practice 2: explain the deciding evidence

Changed-case practice: Add shared bias₂ after each patch sum. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Feature map becomes $[-2, -2]$.

3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Feature map becomes $\begin{bmatrix} -2 & -2 \end{bmatrix}$. Bias adds 2 to each output; it does not change patch products or output shape.

Repair a missing building block

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training and validation curves (ai-5/15). Goal: Compute comparable training and validation losses at saved checkpoints. Short practice: Can the original epoch 3 training loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
